Fiscal Year:	FY 2017	**Task Last Updated:**	FY 11/14/2017
PI Name:	Turek, Fred W Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC) Project: Metagenomic Sequencing of the Bacteriome in GI Tract of Twin Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/14/2014	**End Date:**	12/31/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2018 per NSSC information (Ed., 5/25/18) NOTE: End date changed to 4/13/2018 per NSSC information (Ed., 6/29/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vitaterna, Martha Ph.D. (Northwestern University)		
Grant/Contract No.:	NNX14AH26G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	High throughput DNA/RNA-based technology has led to a revolution in our understanding of the complexity and diversity of the microbiota communities that inhabit many parts of the human body, including those in the gastrointestinal tract (GI). Disruption of normal bacterial communities in the intestine (i.e., dysbiosis) has been linked to several chronic inflammatory diseases and pathological conditions. The complex ecological microbiology community that inhabits the human GI tract influences normal physiology and behavior and the susceptibility to disease. Despite the clear importance of the GI microbiota for maintaining overall health and influencing disease state, the effects of the spaceflight environment on the human microbiome remain unknown. Thus, it is imperative that studies be carried out on long-term missions in space so that any adverse changes can be identified and countermeasures can be employed to insure the safety and health of our astronauts on extended spaceflight missions. This NASA Human Exploration Research Opportunities (HERO) project is centered around sequencing of the microbiome of twin astronauts. It is also complementary to other biomolecular research projects which seek to integrate -omics-based studies on the effects of spaceflight and will involve data-sharing and data analyses with the other Twins research teams.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The spaceflight environment is both unusual and stressful in ways that challenge gastrointestinal and immune function. However, adverse effects of stress on gastrointestinal (GI) and immune function are present in life on Earth as well. The microbiome of the GI tract is generally understood to impact both GI and immune function, but not yet to a degree to which predictions can be made of what microbiome changes may occur in response to stress, or what microbiome changes may predict GI or other health outcomes. By examining the microbiome changes in the context of the other twins studies -omics data, we hope to enhance our understanding of the microbiome's role in shaping responses to both the spaceflight environment and other, Earth-bound stresses.
Task Progress:	The microorganisms (bacteria, viruses, yeast, fungi) present in the GI tract were examined by sequencing DNA isolated from stool samples. The diversity, abundance, complexity, and identities of the microbial communities found have been analyzed. Samples from monozygotic ("identical") twin subjects have been obtained at intervals before, during, and after one twin spent nearly one year onboard the International Space Station. To avoid "batch effects" (introduction of variability between samples processed and analyzed in different batches) we waited until all samples were collected before beginning analysis. Genomic DNA libraries for shotgun metagenomic sequencing were prepared from genomic DNA extracted from fecal swabs. Sheared libraries were assessed for quality and size distribution and 400-600 bp fragments were selected for paired-end Illumina sequencing. Raw sequence data was annotated using the high-throughput search algorithm DIAMOND coupled with the software package MEGAN (MEtaGenome ANalyzer). In addition, the tool SUPER-FOCUS (Subsystems Profile by database Reduction using FOCUS) was used to perform faster functional profiling of large volumes of unannotated metagenomic data. Differential analyses of metagenomic sequence data obtained for samples at different dates and between individuals were performed. Work is now underway to integrate analysis of these results with those from other Twins Study teams.
Bibliography Type:	Description: (Last Updated: 07/16/2025)
Abstracts for Journals and Proceedings	Turek FW, Vitaterna MH, Jiang P, Keshavarzian A, Green SJ. "Metagenomic sequencing of the bacteriome in GI tract of twin astronauts on ground and on one-year ISS mission." 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017